

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

BACKWARD CHAINING

Aim:

To implement the Backward Chaining algorithm in Prolog.

Objective:

To prove a given goal by recursively checking the conditions required to establish it.

Algorithm:

1. Start.
2. Specify the goal.
3. **Check whether the goal is an existing fact.**
4. **If yes, the goal is proved.**
5. **Otherwise, find a rule whose conclusion matches the goal.**
6. **Treat the rule's conditions as subgoals.**
7. **Recursively prove all subgoals.**
8. **If all subgoals are proved, the goal is proved.**
9. **Otherwise, backtrack and try another rule.**
10. Stop.

Flowchart:

START

| v

Set Goal

| v

Is Goal a Fact?

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Yes No

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Proved Find Rule

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Rule Found?

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No Yes

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Fail Create Subgoals

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Prove Subgoals

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All True?

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Yes

No

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| **Proved Backtrack**

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STOP

Prolog Program:

% Backward Chaining

fact(fever). fact(cough).

rule(flu, [fever, cough]). rule(doctor_visit, [flu]). rule(treatment, [doctor_visit]).

backward(Goal) :-prove(Goal, []).

prove(Goal, _) :-fact(Goal),

write('Fact found: '), write(Goal),

nl.

prove(Goal, Visited) :-

\+ fact(Goal),

\+ member(Goal, Visited),

rule(Goal, Conditions),

write('Trying to prove: '), write(Goal),

nl,

prove_all(Conditions, [Goal|Visited]).

prove_all([], _).

prove_all([H|T], Visited) :-prove(H, Visited),

prove_all(T, Visited).

Query

?- backward(treatment). Output

Trying to prove: treatment Trying to prove: doctor_visit Trying to prove: flu

Fact found: fever Fact found: cough true.

Another query:

?- backward(pneumonia).

Output:

Result:

Thus, the Backward Chaining algorithm is successfully implemented to prove a goal by recursively satisfying its required conditions.